

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	18 JUNE 2026
Team ID	INDIVIDUAL PROJECT
Project Name	ShopEZ: E-commerce Application
Maximum Marks	4 Marks

Technical Architecture:

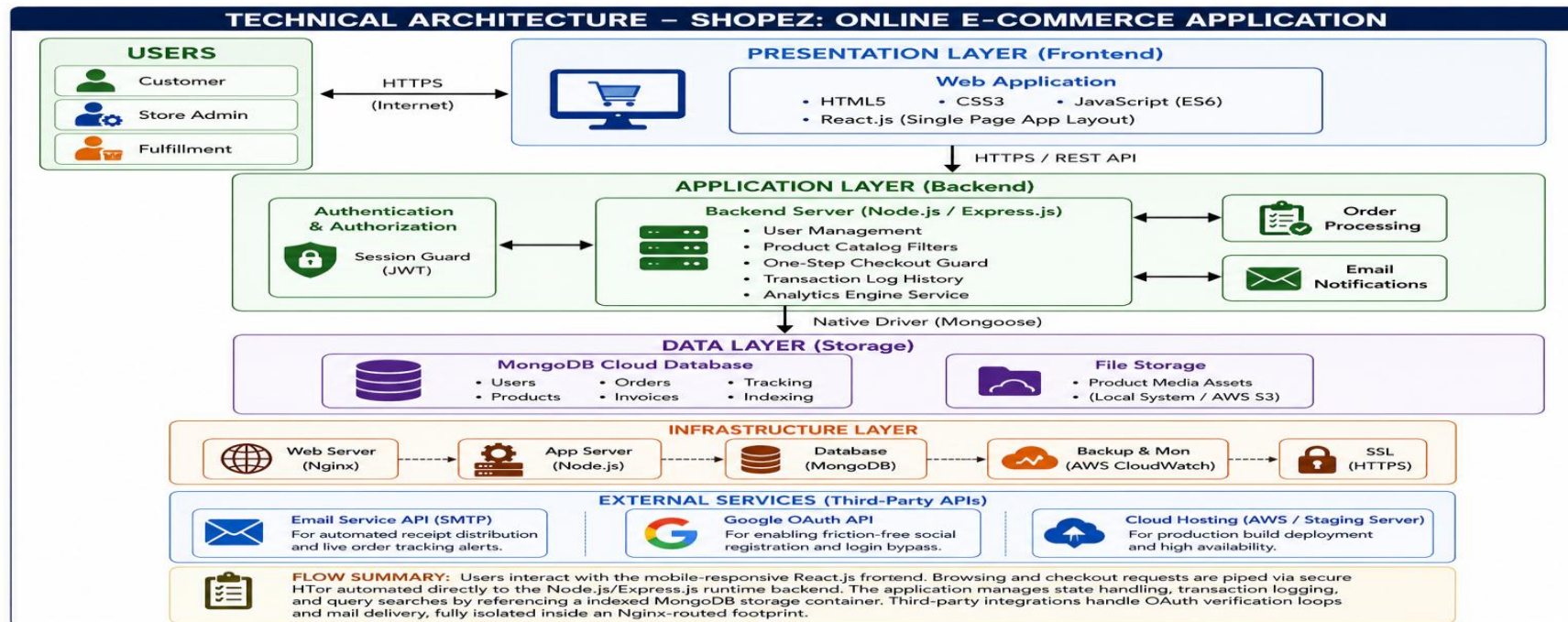


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Responsive interface through which consumers browse products, use search filters, and checkout.	HTML5, CSS3, JavaScript (ES6+), React.js
2	Application Logic-1	User registration, login processing, and secure profile management.	Node.js, Express.js
3	Application Logic-2	Product discovery, compound search filtering, and catalog management.	Node.js, Express.js Backend API
4	Application Logic-3	Single-step secure checkout processing and dynamic order tracking updates.	Node.js, Express.js API
5	Database	Core data store for user records, indexed product catalogs, and transaction documents.	MongoDB (NoSQL) / Mongoose
6	Cloud Database	Production database hosting, automated cluster backups, and scalable sharding.	MongoDB Atlas

S.No	Component	Description	Technology
7	File Storage	Hosts static multi-media digital assets, including product images and shop banners.	Local File System / AWS S3
8	External API-1	Email transaction service for processing order placement confirmations and shipping alerts.	SMTP Email API (e.g., Nodemailer, SendGrid)
9	External API-2	Fast user authentication alternative through social media marketplace integrations.	Google OAuth API
10	Authentication Service	Secures application path logic and routes using stateless verification guards.	JSON Web Tokens (JWT)
11	Web Server	Manages inbound HTTPS server client interactions, request routing, and load balancing.	Nginx
12	Infrastructure	Global application production layer provisioning and codebase micro-deployment.	AWS Cloud / Local Server Staging
13	Monitoring Service	Active application log inspection, database latency checking, and system health tracking.	AWS CloudWatch
14	Machine Learning Model	Personalized customer product recommendation engine.	Not applicable in current version (N/A)

Table-2: Application Characteristics

Characteristics	Description	Technology
1. Open-Source Frameworks	Modern architectural open-source frameworks chosen to deploy optimized frontend layers and rapid async backend environments.	React.js, Node.js, Express.js
2. Security Implementations	Multi-layer data masking incorporating stateless authorization tokens, robust cryptographic account hashing, and SSL network channels.	JWT Authentication, bcrypt Hashing, SSL Certificate (HTTPS)

Characteristics	Description	Technology
3. Scalable Architecture	Decoupled client-server design capable of processing growing volumes of platform consumers, inventories, and checkouts smoothly.	3-Tier Architecture (Presentation, Logic, Data)
4. Availability	High-availability target system deployment strategies integrating continuous monitoring routines to guarantee maximum shopping uptime.	AWS Cloud Hosting, CloudWatch Monitoring
5. Performance	Ultra-responsive performance metrics prioritizing rapid data queries, under 200ms backend latencies, and lag-free UI state updates.	MongoDB Query Indexing, REST API Caching, Nginx Routing